KCH Bronchiolitis Pathway Overview
Bronchiolitis Criteria and Respiratory Score

Viral Bronchiolitis is a clinical diagnosis that can be made by history and physical alone

Diagnosis of Viral Bronchiolitis
- Age ≤24 months
- Increased work of breathing
- Persistent cough
- Feeding difficulties
- Wheezing (+/-)
- Rapid shallow respirations
- Fever (+/-)
- Apnea (+/-)

Inclusion Criteria
- Age ≤24 months old
- Prematurity
- Suspected viral bronchiolitis by history, exam or viral testing

Exclusion Criteria

Cyanotic Congenital Heart Disease
Congenital Heart Disease requiring baseline medications
Unrepaired Anatomic airway defects
Neuromuscular disease
Primary or acquired immunodeficiency
Tracheostomy dependent

KCH Modified Wang Bronchiolitis Severity Scoring

	0	1	2	3
RR **<1 month** **1 mo-12 mo** **13 mo-24 mo**	 <50 <40 <30	 50-60 40-50 31-45	 61-70 51-60 46-60	 >70 >60 >50
Wheeze	None	Terminal Expiratory	Only with stethoscope or throughout expiration	Inspiratory and Expiratory OR audible without stethoscope
Retractions	None	Intercostal only	Tracheosternal	Severe with nasal flaring
General Condition	No issues	PO feeding reduced from normal but adequate	Taking PO feeds but requiring NG or IV fluid	IV or NG fluids only No PO feeding Lethargic

Mild Disease (Score 3-6)

Q6h & PRN (all patients)
- Score
- Suction
- Score

Q6h & PRN (brochodilator responders)
- Score
- Suction
- Albuterol 2.5mg OR MDI 4 puffs
- Score

All patients
- O2 to keep SpO2 ≥90% while awake (88% while asleep)
- Saline drops
- Suction with neosucker
- Patient education
- Discharge criteria assessment

Moderate Disease (Score 7-9)

- **Q4h & PRN (all patients)**
- Score
- Suction
- Score

Q4h & PRN (bronchodilator responders)
- Score
- Suction
- Albuterol 2.5mg OR MDI 4 puffs
- Score

All patients
- Consider HFNC trial
- Consider hypertonic saline
- O2 to keep SpO2 ≥90% while awake (≥88% while asleep)
- Saline drops
- Suction with neosucker
- Parent education
- Discharge criteria assessment

Severe Disease (Score 10-12)

Q2h & PRN (all patients)
- Score
- Suction
- Score

Q2h & PRN (bronchodilator responders)
- Score
- Suction
- Albuterol 2.5mg OR MDI 4 puffs
- Score

All patients
- Consider HFNC
- Consider hypertonic saline
- Consider PICU transfer
- O2 to keep SpO2 ≥90% (≥88% while asleep)
- Saline drops
- Nasopharyngeal suction
- Parent Education

KCH Bronchiolitis Pathway
PED Initial Evaluation

Treatments not routinely recommended
- Acute bronchodilators
- Racemic epinephrine
- Systemic corticosteroids
- Combination inhalers

Initial Assessment
History and Physical
Initial Score
Place in contact/droplet precautions
Suction with NeoSucker

→ **Allow 10 minutes for recovery**

→ **Re- Score patient**
Supplemental O2 to keep SpO2 ≥90%
Assess need for IV fluids
Fill out laminated card and place into room—update until patient disposition finalized

→ **Score>3**

NO → **No bronchodilator trial**
Attempt PO trial
Proceed with family education on Nasal irrigation and bulb suction

YES → **Consider One time Inhaled Bronchodilator Trial IF:**
Respiratory score is ≥9 AFTER suction AND at increased risk of reactive airway disease
See Bronchodilator Trial Algorithm

Bronchodilator RESPONDER
Albuterol as indicated

Bronchodilator NON-RESPONDER
No bronchodilators

→ **Respiratory Score q30 min**
Nasal suction q30 min
At 60 min mark, rescore

→ **60 min Respiratory Score**

≥7 or Attending discretion → **Admit to KCH**
Consider trial of HFNC
(See HFNC pathway)

<7 → **Consider discharge home if other admission criteria are not met**

Inpatient Admit Criteria
- **Respiratory score ≥ 7**
- ***Consider admission for lower Respiratory scores at judgment of ED physician***
- **Persistent hypoxemia (SpO2<90%) on RA or on Supplemental oxygen**
- **Hx Apnea**
- **Dehyrdation or inability to take PO**
- **HFNC implemented with clinical improvement**
- **Consider admission for infants <2months of age, especially if they have co-morbid conditions**

PICU Admit Criteria
- **Score >10**
- **Apnea with bradycardia and cyanosis**
- **Ill appearing (poor perfusion)**
- **Respiratory failure**
- **Failed HFNC trial defined as deteriorating or not improving**
- **Hx of Congenital heart disease (CHP service)**

ED Discharge Criteria
- **RS<7**
- **SpO2 ≥90% on room air**
- **Feeding well**
- **Parents comfortable with nasal irrigation and saline suction**
- **Followup evaluation can be arranged**

KCH Bronchiolitis Pathway
PED HFNC Initiation

Prior to HFNC initiation
- Call respiratory therapist
- Respiratory score→Suction→Score again, done jointly by ED Resident and RT
- MD Orders:
  - -Place IV at PEM provider discretion
  - -consider NS bolus(10-20ml/kg)
  - - NPO
  - -NG/OG tube for gastric decompression at discretion of PED team
  - -Continuous Cardiorespiratory and SpO2 monitoring

ED ATTENDING SHOULD ALWAYS BE NOTIFIED WHEN STARTING A PATIENT ON HFNC!

HFNC Inclusion criteria
Any patient ≤24 months old with suspected viral bronchiolitis
AND EITHER OF:
1. Respiratory score ≥7 with increased work of breathing
OR:
2. Hypoxia refractory to NC or Mask O2

Initiate HFNC
- 2L/kg/min
- 50% FiO2
- Respiratory score q30 min

Although all patients enter the Pathway at 2L/kg/min, HFNC needs to be titrated up or down to effect. Frequent reassessment is critical to determining effect of HFNC

Signs of clinical improvement
- Improving respiratory score
- Lower RR (still within normal limits for age)
- Lower HR (still within normal limits for age)

Huddle at 60 minutes: Respiratory Score on Patient PED RN, PED RT and PED provider evaluate jointly

Worsening →

Trial Failure
- Arrange for PICU admission
- OK to increase flow or increase FiO2 pending Transfer

PICU Admission takes patient off Pathway. Patients admitted to PICU should have expedited transfer out of ED

Improved or unchanged →

Trial Success
- Admit to pediatric hospitalist service
- Follow inpatient pathway
- Consider PICU admission if score≥10, even if improving
- Score, suction, score and vital signs q1h
- Maintain SpO2>90%

PICU admit criteria
- Failed HFNC trial
- Apnea>20 seconds or >20 sec with cardiopulmonary compromise
- Hypoxia despite 50% FiO2
- Altered mental status
- Poor perfusion (cool extremities, cap refill >3seconds)

KCH Bronchiolitis Pathway
Inpatient Management

Resident notification
Verbal: With any increase in severity category
Verbal: When O2 increase is required
Verbal: to suggest albuterol trial for patients who meet criteria
Text: When d/c goals are met

RT and resident assess and score on Admission
Saline & suction
Allow 10 min recovery
Fill out laminated card; place in room

Goal:
Score, suction, score completed within 30 minutes of arrival to inpatient ward

Is score >3?

NO

No Bronchodilators, steroids or antibiotics unless discussed with the attending pediatric hospitalist
Family education on nasal irrigation and bulb suction
NC O2 to keep SpO2 ≥90%

YES

IF NOT done in the PED, consider one time Bronchodilator Trial IF:
Respiratory score is ≥9 AFTER suction AND at increased risk of reactive airway disease
See Bronchodilator Trial Algorithm

Responder?

YES

Saline irrigation and Suction before Treatments
Family education
Include MDI use when applicable
Score all new admits 20 minutes after bronchodilator trial
Repeat Score in 2 hours

NO

Saline irrigation and Suction
Family education
Remove all bronchodilator orders in Epic
Repeat Score and Suction then score again in 2 hours

Mild (Score 3-6)
- Score, Suction Score q6h &PRN
- Hydration (PO/NG/IV)

Moderate (Score 7-9)
- Score, suction score q4h & PRN
- Hydration (PO/NG/IV)
- Consider HFNC
- Bronchodilators if responsive

Severe (Score 10-12)
- Score, suction score q2h &PRN
- Consider HFNC
- Consider 3% saline nebs
- Consider PICU admission

Avoid Routine Use of
Viral testing, Chest XR, CBC, lytes, VBG, UA, Blood or urine culture, Albuterol and epinephrine, Corticosteroids, Antibiotics, Antivirals, Chest physiotherapy, Hypertonic saline, Deep suctioning

Inpatient Discharge Considerations
- Respiratory Score <6 for >6 hours
- Nasal irrigation and bulb suction only
- No O2 requirement ≥6 hours
- SpO2 ≥90% awake, ≥88% asleep
- No further apnea in ≥ 24 hours
- Adequate PO intake
- Adequate Urine output
- Parent teaching completed and comfort assured

KCH Bronchiolitis Pathway
Inpatient HFNC Initiation and Maintenance

Phase 1: Initiation

Prior to HFNC Initiation:
Score, Suction, Score
Consider ONE-TIME bronchodilator if not already done and patient meets previous criteria
Resident to Notify Attending Hospitalist

HFNC Inclusion criteria
Any patient ≤24 months old with suspected viral bronchiolitis
AND EITHER:
1. Respiraotry score ≥7 with increased work of breathing **OR**
2. Hypoxia refractory to NC of Mask O2

Initiate HFNC
2L/kg/min
FiO2 50%
VS q30 min

MD/DO to order:
- Place PIV
- NPO for initiation
- NG/OG tube for gastric decompression at hospitalist team discretion
- Consider NS bolus (10-20 ml/kg)
- Continuous Cardiorespiratory and SpO2 monitoring

Huddle At 90 minutes
(RN, RT, RN Team lead, Resident)
Joint scoring of patient

Improved or unchanged?

YES → **Trial success**
Proceed to Phase 2

NO → **Trial Failure (worsening)**
- Emergent transfer to PICU
- Maintain current HFNC settings

CHANGES TO FLOW RATE OR FIO2 OUTSIDE OF PICU CAN BE MADE IN COLLABORATION BETWEEN ATTENDING AND RT

Phase 2: Maintenance & Weaning

Maintain HFNC support for 4 hours
Transfer to PICU for any clinical deterioration
Score q1h x 2 hours, then again at 4 hours
Determine safest hydration/nutrition method (PO/NG/IV)
Goal oxygen SpO2 ≥90; wean FiO2 for SpO2 ≥95

Complete 4 hours of observation
If stable, consider weaning FLOW
See HFNC Weaning Algorithm

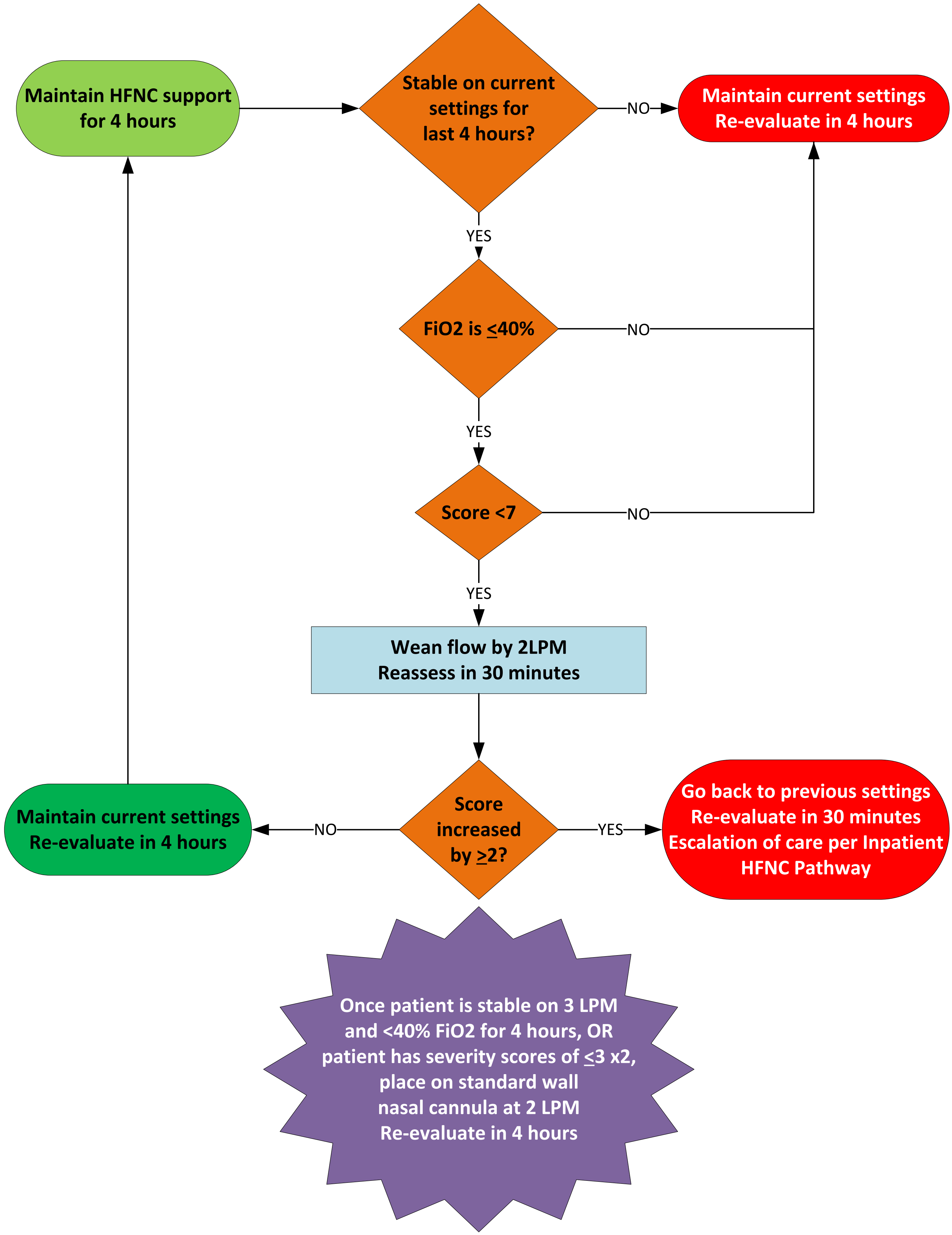
Considerations for FiO2 weaning
- OK to wean FiO2 after 2 hour observation period
- Score q2h x 2 hours, then q4h
- Goal SpO2 ≥90%
- Goal SpO2 ≥88% in patients with chronic oxygen dependence
- Wean FiO2 if SaO2 ≥95%

Weaning FiO2
- Wean FiO2 in 5-10% increments based on MD/DO order
- Can be done by RT for SpO2 goals listed above
- Reassess 5-10 minutes after wean
- May wean more aggressively per MD/DO order if tolerating wean
- Goal FiO2 <30%

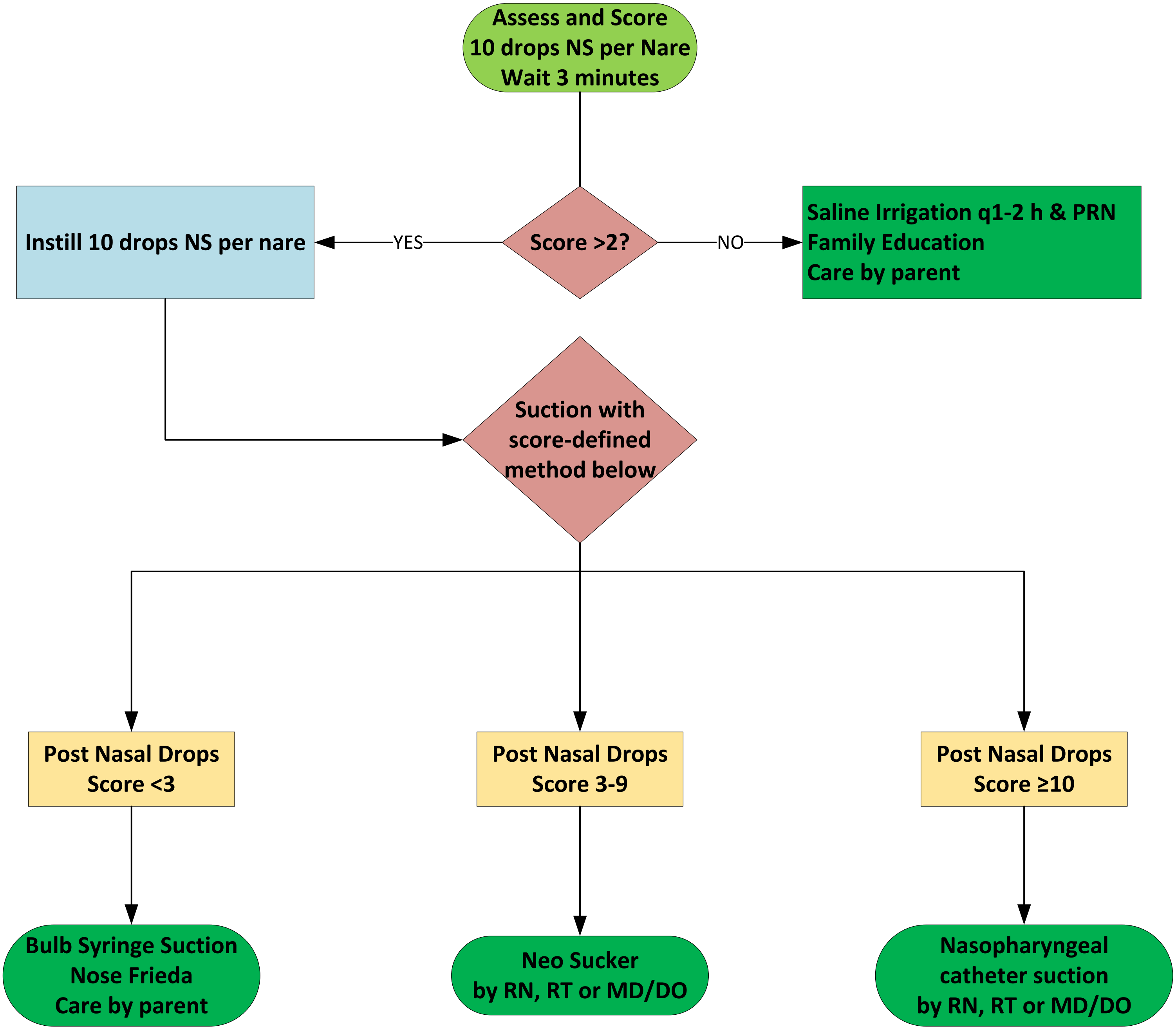
PICU transfer criteria
- Failed HFNC trial
- Apnea>20 seconds or <20 sec with cardiopulmonary compromise
- Hypoxia despite 50% FiO2 (SpO2<90%)
- Altered mental status
- Poor perfusion (cool extremities, cap refill >3seconds

Signs of clinical improvement
- Improving respiratory score
- Lower RR (still within normal limits for age)
- Lower HR (still within normal limits for age)

KCH Bronchiolitis Pathway
Inpatient HFNC Initiation and Maintenance

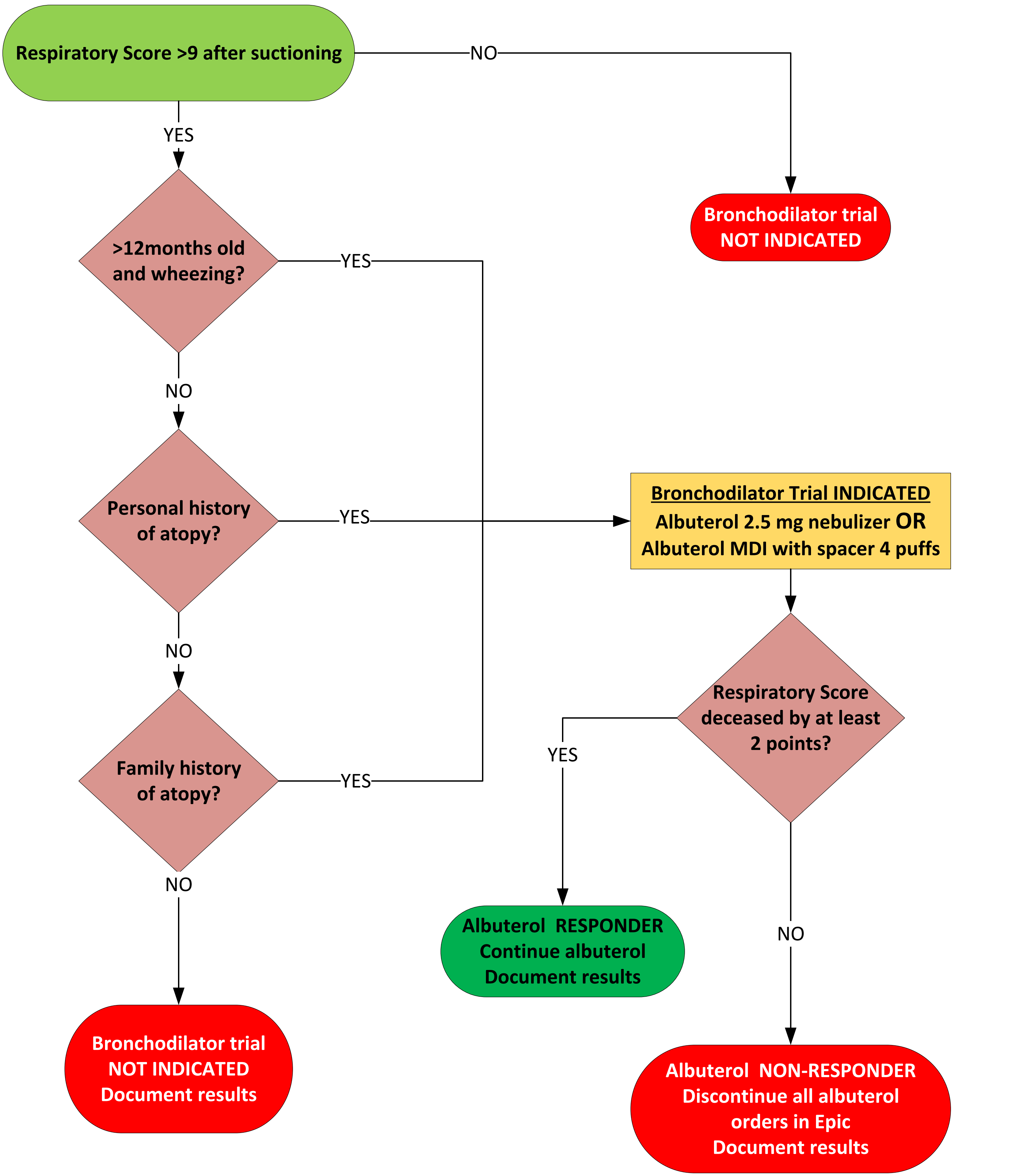


KCH Bronchiolitis Pathway
Suction Algorithm



Rescore 10 minutes after suctioning

KCH Bronchiolitis Pathway
Bronchodilator Trial Algorithm



My Name is : ___________________________

I had an Albuterol trial on ___________ at ______
I DID/DID not respond to Albuterol.

My last pre suction score: _________

My last post suction score was: _________

If indicated:
My post neb score was: _________

The last things my RT did for me were:
______ O2 Wean (Pass / Fail)
______ Increase O2
______ Suction
______ Albuterol neb

ED version: In room scoring

KCH Modified Wang Bronchiolitis Severity Scoring				
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General Condition	No issues	PO feeding reduced from normal but adequate	Taking PO feeds but requiring NG or IV fluid	IV or NG fluids only No PO feeding Lethargic

	Day Shift	**Night Shift**
Date / Time Pre / Post suction/ (Post neb if indicated)		
Date / Time Pre / Post suction (Post neb if indicated)		
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Date / Time Pre / Post suction (Post neb if indicated)		
Date / Time Pre / Post suction (Post neb if indicated)		

My Name is : ___________________________

I had an Albuterol trial on ___________ at _______
I DID/DID not respond to Albuterol.

My RT is _____________.

My RT checked on me last at ___________ on _________.

My last pre suction score: __________

My last post suction score was: __________

My RT will be back in about ________ hours.

The last things my RT did for me were:
______ O2 Wean (Pass / Fail)
______ Increase O2
______ Suction
______ Albuterol neb

KCH version: In room scoring

KCH Modified Wang Bronchiolitis Severity Scoring

	0	1	2	3
RR				
<1 month	<50	50-60	61-70	>70
1 mo-12 mo	<40	40-50	51-60	>60
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Resources

1. Ralston SL, Lieberthal AS, Meissner HC, Alverson BK, Baley JE, Gadomski AM, et al. Clinical practice guideline: the diagnosis, management, and prevention of bronchiolitis. *Pediatrics*. 2014;134(5):e1474-1502. https://www.ncbi.nlm.nih.gov/pubmed/25349312

2. Dalziel SR, Haskell L, O'Brien S, Borland ML, Plint AC, Babl FE, et al. Bronchiolitis. *Lancet*. 2022;400(10349):392-406. https://www.ncbi.nlm.nih.gov/pubmed/35785792

3. AAP. Respiratory Syncytial Virus. In: AAP, editor. *Red Book*. 32 ed. Elk Grove: AAP; 2021. p. 628-636.

4. Kou M, Hwang V, Ramkellawan N. Bronchiolitis: From Practice Guideline to Clinical Practice. *Emerg Med Clin North Am*. 2018;36(2):275-286. https://www.ncbi.nlm.nih.gov/pubmed/29622322

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The version replaces the original version released in 2019.